$$I_E = \varepsilon I_{0,th} = 1,37 \cdot 10^3 \frac{W}{cm^2}. \quad (24)$$

3.13 (19) (20) . (19)

$$\bar{n} = \frac{1}{KT}, \quad (25)$$

(20)

$$I_G = \frac{I_0 \sigma_A - \frac{1}{\tau}\bar{n}}{2\sigma_E \bar{n}}. \quad (26)$$

$$I_0 = \eta I_{0,th} = \eta \frac{\bar{n}}{\tau \sigma_A} \quad (27)$$

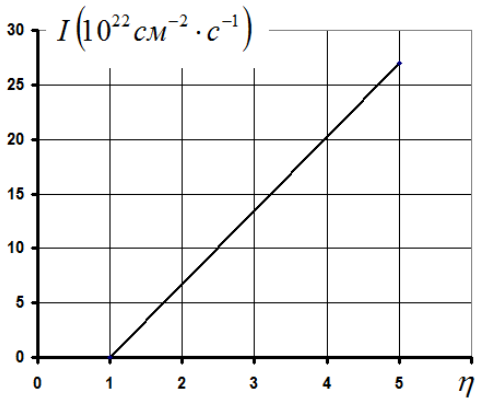
(23)

$$I_G = \frac{\eta \frac{\bar{n}}{\tau\sigma_A}\sigma_A - \frac{1}{\tau}\bar{n}}{2\sigma_E \bar{n}} = \frac{\eta - 1}{2\tau\sigma_E}. \quad (28)$$

$$I = (1-\rho) I_G = \frac{(1-\rho)}{2\tau\sigma_E}(\eta - 1) = E(\eta - 1), \quad (29)$$

$$E = \frac{1-\rho}{2\tau\sigma_E} = 5,41 \cdot 10^{22} cm^{-2} \cdot s^{-1}. \quad (30)$$

(29) .



3.14 ,

$$N_A = \eta I_{0,th} \sigma_A \gamma S l. \quad (31)$$

,

$$N_E = E(\eta - 1)S. \quad (32)$$

$$f = \frac{N_E}{N_A} = \frac{E(\eta - 1)}{(I_0)_{tr} \sigma_A (\gamma l)}. \quad (33)$$

$$f = \frac{\eta - 1}{\eta}. \quad (34)$$